TELECOM · CHURN ANALYSIS

When Half Your Customers Walk Out.

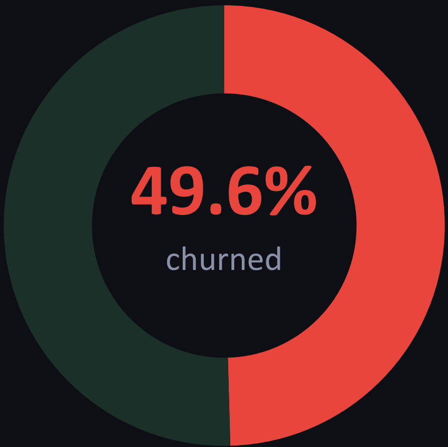
A data-driven churn analysis across 100,000 telecom customers.

49.6%	$28-35M	2×
Churn Rate	Revenue at Risk	Industry Average

THE PROBLEM

Nearly half of all customers churned.

2× the industry average of 20–25%



49,562
Customers Lost
Out of 100,000 total

50,438
Customers Stayed
50.4% retention

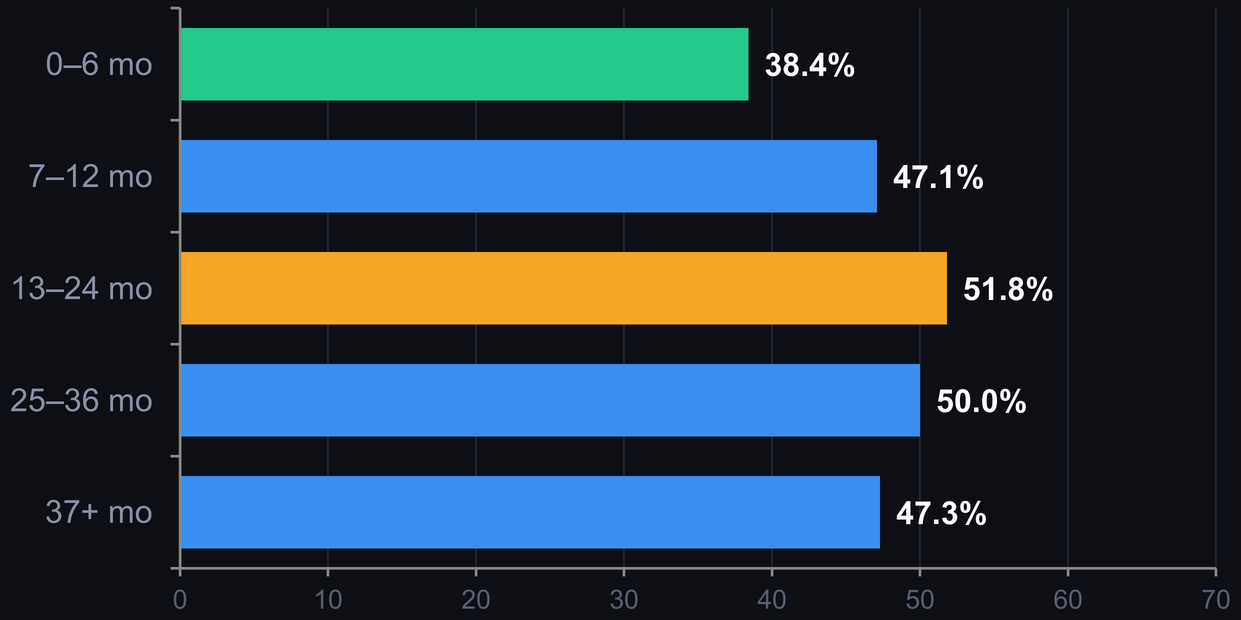
2×
Industry Average
Benchmark: 20–25%

KEY INSIGHT At 49.6%, churn is a structural revenue problem — not an outlier event.

DISCOVERY

Months 13–24 is the danger zone.

Churn peaks once the initial deal honeymoon ends.



KEY INSIGHT Past the deal, before loyalty — that gap is the retention window.

ROOT CAUSE

Older devices.
Higher churn.

A consistent, monotonic relationship across all device age groups.

41.8%	47.0%	53.8%	57.9%
< 6 mo	6–12 mo	1–2 yr	2+ yr
n=18,161	n=36,356	n=34,505	n=10,811

+16.1 pp churn gap between newest and oldest devices

eqpdays × hnd_price correlation

r = –0.48 (correlated features)

Negative eqpdays excluded (data quality)

133 records removed before analysis

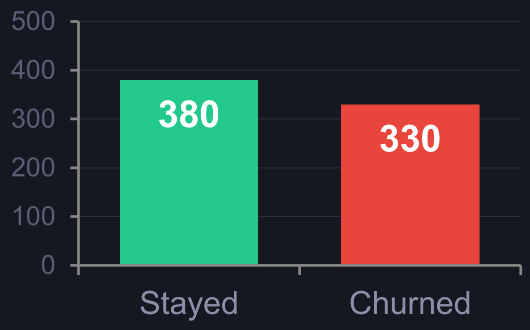
KEY INSIGHT Device age and handset price are correlated (r=–0.48) — likely the same underlying signal.

BEHAVIORAL SIGNAL

Customers go silent before they leave.

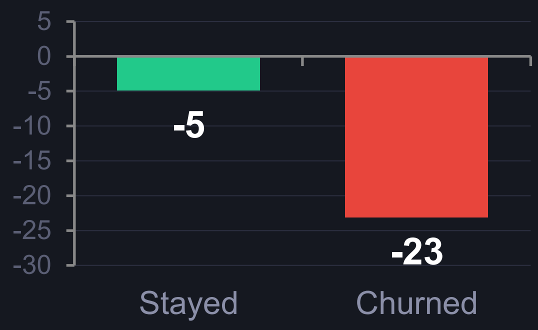
Usage level and usage trend both separate churners from loyal customers.

MONTHLY MINUTES — MEDIAN



Cohen's d = 0.11 · small effect · $p < 0.001$

USAGE CHANGE VS PRIOR 3 MONTHS



Cohen's d = 0.08 · outliers clipped · $p < 0.001$

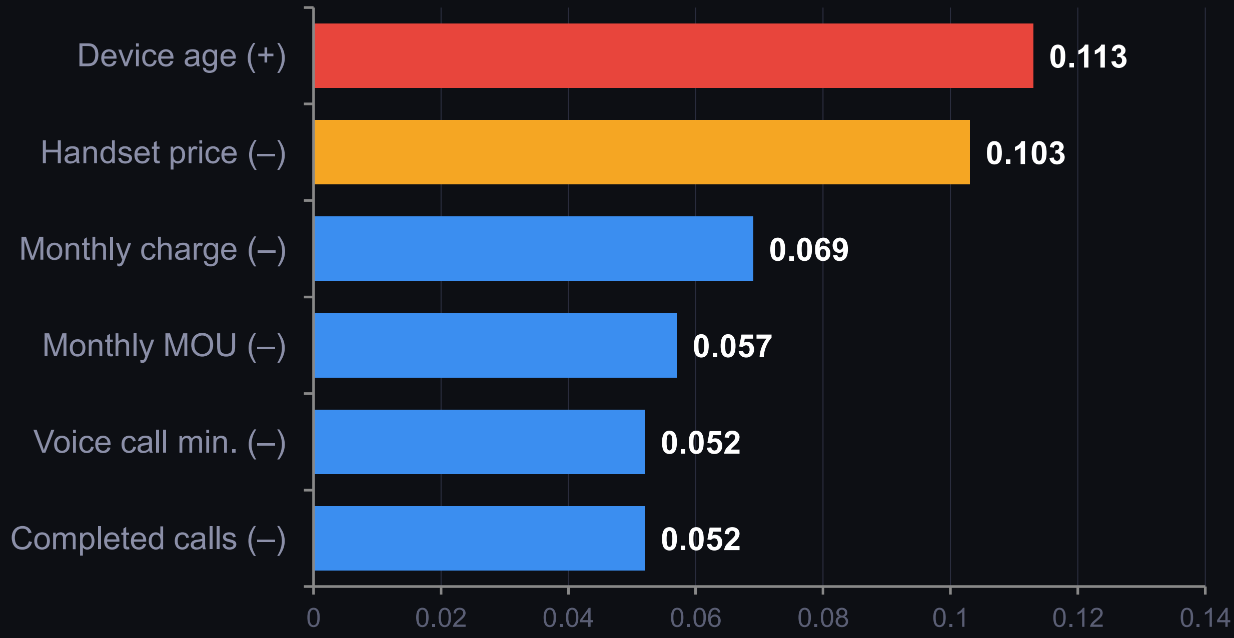
4.7× steeper usage decline in churners (–23.1 min vs –4.9 min / month)

KEY INSIGHT Significant association but small effect — useful as one signal in an ensemble model.

CORRELATION ANALYSIS

Top correlated features.
Not causal drivers.

Point-biserial correlation with churn · All |r| < 0.15



(+) correlated with churn · (–) negatively correlated with churn

KEY INSIGHT All correlations are weak (|r| < 0.15). Non-linear modeling is needed to capture the full signal.

KEY TAKEAWAYS

What the data actually tells us.

01 **49.6% churn rate**

2× industry benchmark — structural revenue problem, not an outlier.

02 **Months 13–24**

Peak churn window. +2.2 pp above average. χ^2=236, statistically significant.

03 **Older devices**

Linear rise: 41.8% (new) → 57.9% (2+ yr). A 16.1 pp gap.

04 **Usage decline signal**

Churners drop usage 4.7× faster. Small but consistent and detectable.

Revenue at risk: **$28.2M–$34.6M / yr** · Next: Feature Engineering + ML Modeling